Práctica dos: Sistema cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

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Información general



Nombre del alumno: Mildred Angel Torales

Número de control: 23212731

Correo institucional: l23212731**@tectijuana.edu.mx**

Asignatura: **Modelado de Sistemas Fisiológicos**

Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

Datos de la simulación

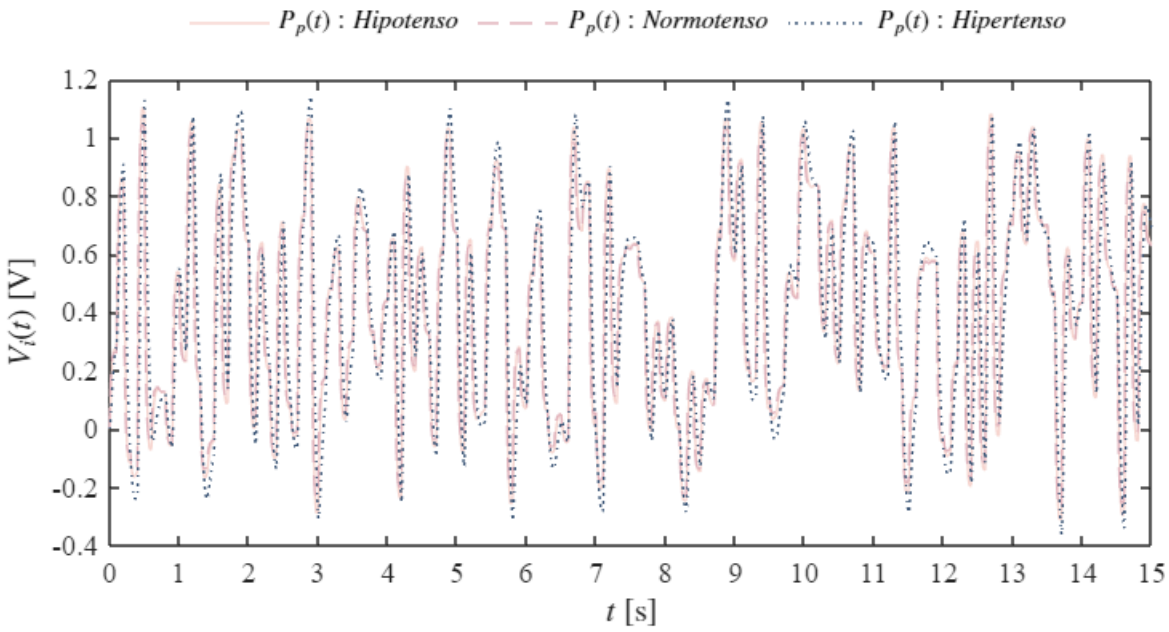
```
clc; clear; close all; warning('off','all')
open_system('Practica2_MSF1_MAT')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
```

```
parameters.MaxStep = '1e-2';
x= sim('Practica2_MSF1_MAT',parameters);

t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data,x.Pa,t, 'previous');
writematrix(x.Pa,'signal.xlsx');
```

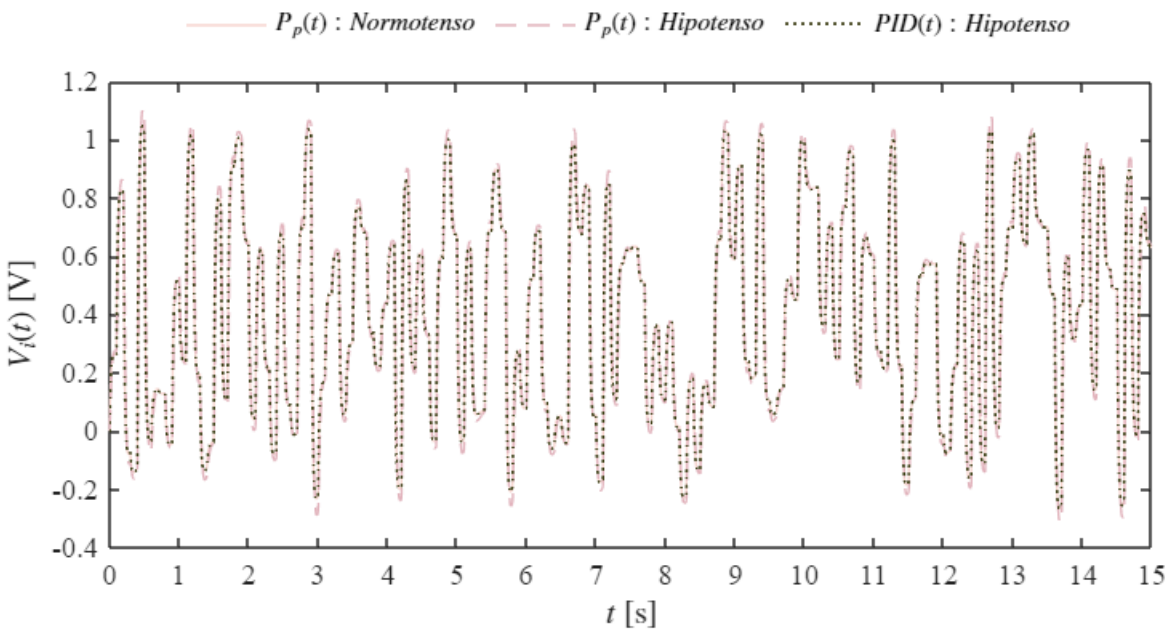
Sistema normotenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH,'Normotenso')
```



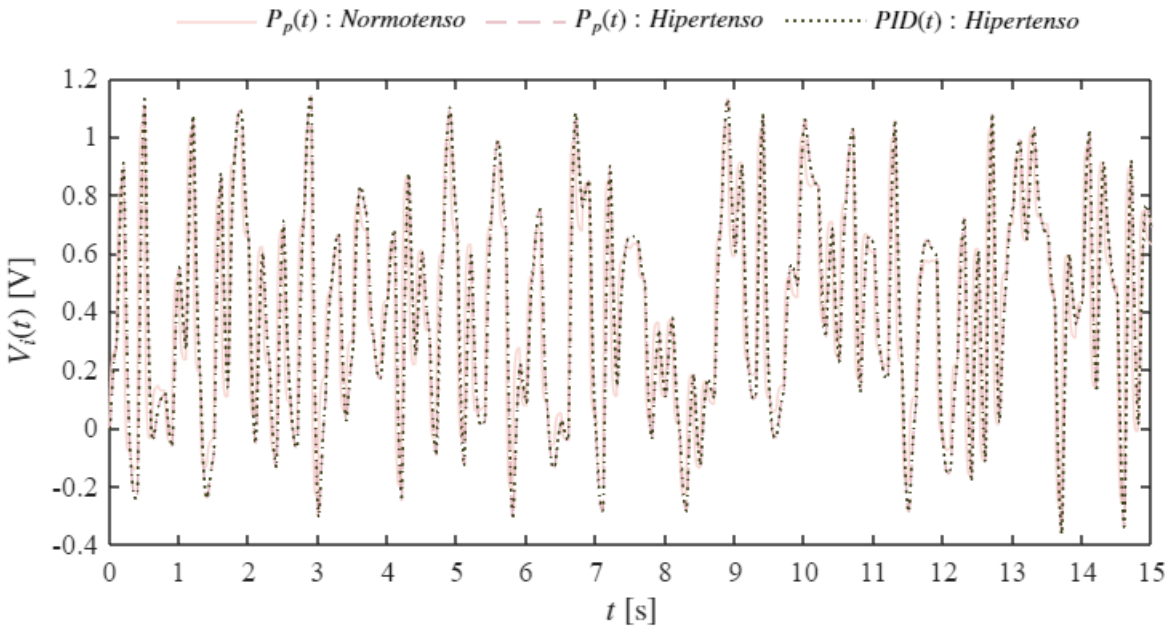
Sistema hipotenso

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'Hipotenso')
```



Sistema Hipertenso

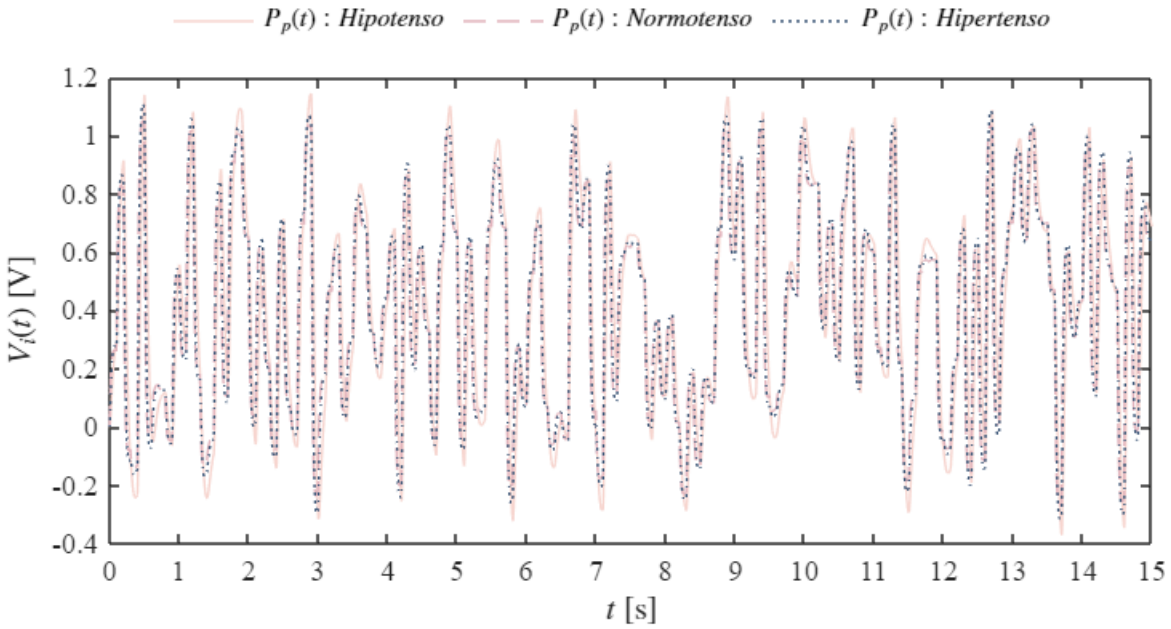
```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'Hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.01,0.001,1.00,3.500];
```

```
par_normo = [0.01,0.001,1.00,1.500];
par_hiper = [0.01,0.001,1.00,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'odes');
```




```
function Pp = cardio(par,Pa,t)
    Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
    n = round(t(end)/dt);
    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;
    for i = 1:n
        FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))* Pp(i) + Pa(i)/(Z*C))*dt;
    end

end
```

Funcion: Respuesta de simulacion

```
function plotsignals(t,x1,x2,x3,caso)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [247, 214, 208]/255;
    clr2 = [226, 180, 189]/255;
    clr3 = [62, 88, 121]/255;
    clr4 = [65, 67, 27]/255;
```

```
    if caso == "Normotenso"
        plot(t,x1,'-','LineWidth',0.25,'Color',clr1)
        plot(t,x2,'--','LineWidth',0.25,'Color',clr2)
        plot(t,x3,':','LineWidth',0.25,'Color',clr3)
        L =
legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
    elseif caso == "Hipotenso"
        plot(t,x1,'-','LineWidth',0.25,'Color',clr1)
        plot(t,x2,'--','LineWidth',0.25,'Color',clr2)
        plot(t,x3,':','LineWidth',1,'Color',clr4)
        L = legend('$P_p(t):Normotenso$','$P_p(t):Hipotenso$','$PID(t):Hipotenso$');
    elseif caso == "Hipertenso"
        plot(t,x1,'-','LineWidth',0.25,'Color',clr1)
        plot(t,x2,'--','LineWidth',0.25,'Color',clr2)
        plot(t,x3,':','LineWidth',1,'Color',clr4)
        L =
legend('$P_p(t):Normotenso$','$P_p(t):Hipertenso$','$PID(t):Hipertenso$');
    elseif caso == "odes"
        plot(t,x1,'-','LineWidth',0.25,'Color',clr1)
        plot(t,x2,'--','LineWidth',0.25,'Color',clr2)
        plot(t,x3,':','LineWidth',0.25,'Color',clr3)
        L =
legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
    end


set(L,'interpreter','latex','location','northoutside','orientation','horizontal',
'box','off')
    xlabel('$t$ [s]','interpreter','latex')
    ylabel('$V_i(t)$ [V]','interpreter','latex')

    xlim([0 15]); xticks(0:1:15)
    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

end
```